

CANONICAL LESSON

Outbreak Signals Before Alarm

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C21
Notebook	V7_B_N01
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Detect possible outbreaks through governed baselines, thresholds, persistence, corroboration, uncertainty, and authorized public-health escalation.

Learning outcomes

Why this matters

Premature alarms can cause panic, while delayed escalation can allow a genuine outbreak to spread.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

A syndromic signal exceeding a seasonal baseline is corroborated across facilities before epidemiological review.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Add a moving-baseline method.
- 2.
- Test threshold sensitivity.
- 3.
- Define a confirmation workflow.

- 4.
- Explain why public communication cannot be automated from this notebook.

Answer key

- 1.
- Use only prior weeks, preserve seasonality, and avoid contaminating baselines with active events.
- 2.
- Compare verified-event sensitivity, false alerts, delay, workload, and subgroup/geographic effects.
- 3.
- Assign triage, data verification, epidemiological assessment, laboratory/clinical review, authorization, communication, closure, and retrospective review.
- 4.
- Communication carries legal, scientific, behavioural, privacy, and political consequences and requires authorized contextual judgement.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.